IEOR 145/245 · GROUP SIMULATION EXERCISE · GAME PLAYED OCTOBER 20, 2026

30-Customer Assortment Game

We sold all 18 units for the maximum possible revenue of $270, selling the last unit to customer 27 of 30. The assortments came from a dynamic program built on customer-type choice models fitted to the training data. Customers 3–27 were shown exactly what it recommended; customers 1 and 2 were shown {2,3} instead, which cost about $0.13 in expected revenue.

REVENUE **$270** the maximum: 6 units each at $10, $15, $20	SOLD OUT AT **27 / 30** 3 customers to spare
FORECAST BEFORE THE GAME **$268.7** model's expected revenue under our policy	CHANCE OF $270 ON THIS CUSTOMER SEQUENCE **94%** as played; 51% for the myopic greedy policy

The game

- Three products: product 1 at $10, product 2 at $15, product 3 at $20, with 6 units of each and no restocking.
- 30 customers arrive one at a time. Each is type A, B or C, and the type is revealed on arrival.
- For each customer we choose which in-stock products to show. The customer buys at most one of them or walks away.
- Goal: maximize total revenue. Since there are only 18 units for 30 customers, the $270 maximum requires selling every unit.
- Training data: 63,000 past customers with their type, the assortment shown and the purchase. Every type was shown every one of the 7 possible assortments (A 3,300 times each, B 2,100, C 3,600).

What each type buys

The three types behave very differently. The table gives the purchase rate when each product is shown on its own, taken directly from the training data, and the weights of a multinomial logit (MNL) model fitted separately to each type by maximum likelihood (Lecture 14A, no-purchase weight = 1).

TYPE	BUYS PRODUCT 1 ALONE	BUYS PRODUCT 2 ALONE	BUYS PRODUCT 3 ALONE	MNL WEIGHTS V_1, V_2, V_3	IN SHORT
A	18%	16%	**62%**	0.20, 0.19, 1.59	Buys product 3 or nothing
B	49%	**87%**	51%	1.01, 6.95, 1.01	Strongly prefers product 2
C	66%	67%	**68%**	2.01, 2.07, 2.04	Buys whatever is shown

Model fit: across all 7 assortments, predicted probabilities are within 0.021 of the observed frequencies. Likelihood-ratio tests against an unrestricted model give p = 0.47 (A), 0.034 (B) and 0.66 (C); B's result is not significant after a Bonferroni correction for three tests. A single MNL pooled across types is rejected (LR = 17,398, 6 df), so the model is personalized by type.

The policy

Lecture 14B poses this problem (online assortment with finite inventory and personalized choice models) and solves it with a greedy rule: show each customer the assortment with the highest expected revenue *for that customer*. Here greedy shows {2,3} to every type at the start and leaves the $10 product for later, which is a problem because type A rarely buys it.

We instead solve the full problem with a dynamic program. The state is the customer number and the remaining inventory; arriving types are modeled as independent draws with the mix seen in the data (11 A : 7 B : 12 C out of 30). With 30 customers, 7^3 inventory levels and 8 possible assortments, the program is solved exactly in milliseconds.

$$\text{value of showing } S \text{ to a type-}k \text{ customer} = V_{\text{next}}(x) + \Sigma_{i \in S}\, P_k(\text{buy } i \mid S) \cdot (r_i - \Delta_i)$$

Here Δ_i is the **opportunity cost** of product i: how much expected future revenue one unit is worth if kept. Each decision is therefore the Lecture 13 MNL assortment problem with every price r_i replaced by an **adjusted price** $\rho_i = r_i - \Delta_i$. We proved that the optimal set is always the in-stock products with the highest adjusted prices, and checked this, along with the class LP, at all 30,780 possible decision points.

At the start, every unit is likely to sell eventually, so each opportunity cost is close to the full price. The adjusted prices are ρ = (1.20, 0.84, 0.71) for products 1, 2, 3: the cheapest product is the most urgent to sell.

POLICY, EVALUATED BEFORE THE GAME	EXPECTED REVENUE	SELLS ALL 18 UNITS	5TH-PERCENTILE REVENUE
Dynamic program (ours)	**$268.57**	**91.5%**	**$260**
Show all in stock	$267.66	85.8%	$250
Inventory balancing (Golrezaei et al. 2014)	$265.81	76.5%	$250
Myopic greedy (Lecture 14B)	$260.97	56.3%	$230

Expected revenue is computed exactly by policy evaluation, with customer behavior set to the observed frequencies in the training data. Sell-out rates and percentiles come from 50,000 simulated games with common random numbers. Types arrive i.i.d. in the data's mix.

How the game went



bought product 1 bought product 2 bought product 3 no purchase

Projected final revenue as each customer arrived (revenue already earned plus the model's expected revenue from the remaining customers). The strip under the axis shows what each customer bought.

CUSTOMERS 1–3

Opening. The simulator record shows customers 1 (A) and 2 (B) were shown $\{2,3\}$, not the policy's $\{1,2,3\}$ and $\{1,2\}$. The outcomes were what the policy's assortments could also have produced (A bought nothing, B bought product 2), so the inventory and every later state matched the policy's path. Ex ante the two deviations cost $0.05 and $0.08 in expected revenue. From customer 3 on, assortments followed the policy: C saw $\{1,2\}$, holding product 3 back for type A, and bought product 2.

CUSTOMERS 4–17

Save product 2 for type B. Once product 2 dropped to 4 units, type C customers (who buy anything) were steered to $\{1\}$, and later to $\{1,3\}$. The one exception was customer 6, where the three adjusted prices were within $0.50 of each other, so C saw everything and bought product 2. Type B kept seeing product 2 and bought it at customers 8 and 14. At customer 9, the adjusted prices were $\rho = (2.01, 1.22, 1.50)$, so the top-ranked set for C was $\{1,3\}$. Customers 10–13 bought nothing, four in a row, and the projection fell to its low point of $261.0. Then three C customers in a row bought product 1 (customers 15–17).

CUSTOMERS 18–22

Push product 3. With all 6 units of product 3 left and 13 customers to go, product 3 became the most urgent: $\rho = (3.85, 3.55, 6.19)$ at customer 18, so C was shown $\{3\}$ alone and bought it. Type A customers saw $\{1,3\}$, which kept the last unit of product 2 for type B, who bought it at customer 20. Another B bought the last unit of product 1 at customer 22.

CUSTOMERS 23–27

Clear the last units. Only product 3 remained, so everyone saw $\{3\}$. Two B customers passed, then three A customers in a row bought it.

CUSTOMER	TYPE	INVENTORY	SHOWN	P(NO BUY)	OUTCOME	REVENUE	PROJECTED
1	A	(6, 6, 6)	{2,3} (policy: {1,2,3})	0.36	none	$0	$268.7
2	B	(6, 6, 6)	{2,3} (policy: {1,2})	0.11	product 2 ($15)	$15	$268.0
3	C	(6, 5, 6)	{1,2}	0.20	product 2 ($15)	$30	$268.2
4	C	(6, 4, 6)	{1}	0.33	none	$30	$268.3
5	C	(6, 4, 6)	{1}	0.33	product 1 ($10)	$40	$267.5
6	C	(5, 4, 6)	{1,2,3}	0.14	product 2 ($15)	$55	$268.4
7	A	(5, 3, 6)	{1,2,3}	0.34	none	$55	$268.5
8	B	(5, 3, 6)	{1,2,3}	0.10	product 2 ($15)	$70	$267.7
9	C	(5, 2, 6)	{1,3}	0.20	product 1 ($10)	$80	$267.7
10	C	(4, 2, 6)	{1,3}	0.20	none	$80	$268.5
11	C	(4, 2, 6)	{1,3}	0.20	none	$80	$267.6
12	C	(4, 2, 6)	{1,3}	0.20	none	$80	$266.2
13	A	(4, 2, 6)	{1,2,3}	0.34	none	$80	$264.1
14	B	(4, 2, 6)	{1,2,3}	0.10	product 2 ($15)	$95	$261.0
15	C	(4, 1, 6)	{1,3}	0.20	product 1 ($10)	$105	$261.0
16	C	(3, 1, 6)	{1,3}	0.20	product 1 ($10)	$115	$262.5
17	C	(2, 1, 6)	{1,3}	0.20	product 1 ($10)	$125	$263.3
18	C	(1, 1, 6)	{3}	0.33	product 3 ($20)	$145	$263.7
19	A	(1, 1, 5)	{1,3}	0.36	product 3 ($20)	$165	$266.0
20	B	(1, 1, 4)	{1,2,3}	0.10	product 2 ($15)	$180	$267.7
21	A	(1, 0, 4)	{1,3}	0.36	product 3 ($20)	$200	$267.6
22	B	(1, 0, 3)	{1,3}	0.33	product 1 ($10)	$210	$268.8
23	B	(0, 0, 3)	{3}	0.50	none	$210	$269.0
24	B	(0, 0, 3)	{3}	0.50	none	$210	$267.9
25	A	(0, 0, 3)	{3}	0.39	product 3 ($20)	$230	$265.9
26	A	(0, 0, 2)	{3}	0.39	product 3 ($20)	$250	$268.2
27	A	(0, 0, 1)	{3}	0.39	product 3 ($20)	$270	$269.5

Shown is the assortment recorded by the simulator; where it differs from the policy's recommendation, the recommendation is noted. Inventory is the stock of products 1, 2, 3 when the customer arrived; P(no buy) is the model's chance the customer walks away from the assortment shown; revenue is the running total after the customer; projected is revenue so far plus the model's expected revenue from the remaining customers. Source: the simulator's history file (`LIVE-RHR5BR_assortment_simulation_history.csv`), cross-checked against the `play.py` session log. Types, purchases, inventory and revenue agree in both records; the shown assortments agree for customers 3–27.

Policy or luck?

PURCHASES

9 customers bought nothing, against **7.4** expected for the assortments shown. The outcomes ran slightly against us.

ARRIVALS AMONG THE FIRST 27

`8 A, 7 B, 12 C`, against `9.9, 6.3, 10.8` expected. Slightly fewer of the pickiest type (A) and more of the least picky (C) is a small advantage.

To separate the policy from luck, we replayed the same customer types through each policy and computed the exact chance of reaching $270.

POLICY, SAME 27 CUSTOMER TYPES	EXPECTED REVENUE	CHANCE OF $270
As we actually played customers 1–2 shown {2,3}, then the policy	**$269.04**	94.0%
Dynamic program, followed for all 27	$269.09	94.3%
Show all in stock	$268.22	88.4%
Inventory balancing	$266.02	76.6%
Myopic greedy (Lecture 14B)	$259.35	50.7%
Clairvoyant planner knows all 27 types in advance	$269.22	95.1%

Exact values from policy evaluation. Customers 1–27 have the types observed in the game. Customers 28–30 never arrived, so their types are drawn from the data's mix. Purchase behavior uses the training data's observed frequencies; using the fitted MNL instead changes each value by at most $0.15 and 0.5 points.

Facing these customers, the way we played had a 94.0% chance of $270 (94.3% had customers 1–2 also followed the policy). That is within about one point of a planner who knew all 27 customers' types in advance (95.1%). The same customers would have given the greedy rule from class roughly a coin flip (50.7%). The $270 was not guaranteed, but it was the likely outcome of the policy, not a lucky one.

Limitations

- The policy assumes customers in the simulation behave like those in the training data. A single game is too short to test that assumption.
- The arrival process of types was not given. We planned with the data's mix and tested 13 scenarios (shifted mixes and adversarial orders such as all type A last). The policy's largest loss against a planner who knew the true process was $2.79.
- One game is one random draw. Even with the best policy, revenue over a 30-customer game has a standard deviation of about $5.50.

Reproduce: `code/rm_model.py` (estimation, dynamic program, baselines, evaluation), `notebooks/RM_assortment_model.ipynb` (full analysis, proofs and robustness tests), `code/play.py` (the live helper used in this game), `outputs/policy_table.csv` (the policy as a lookup table) and `simulation_results/` (this game's records).